

Souherdya Sarkar

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SUMMARY

Passionate software developer with hands-on experience in building impactful applications that address real-world challenges. Strong interest in Artificial Intelligence and Machine Learning, with a focus on leveraging data-driven solutions to enhance functionality and user experience. Skilled in full-stack development, problem-solving, and continuous learning in emerging technologies.

EDUCATION

Institute of Engineering and management

B.Tech in computer science and engineering ; GPA - 9.7(Aggregate)

Uttarpara Children's Own Home

Higher Secondary

Kolkata, West bengal

August 2023 – May 2027

Uttarpara, West Bengal

Jan. 2021 – Feb 2023

EXPERIENCE

Software Developer Intern

IEMA Research and Development Pvt. Ltd.

May 2025 – Present

On-Site

- Optimized performance of the AI web application **IEMA.ai**, improving load speed and responsiveness by 20%
- Redesigned backend architecture and implemented stronger security protocols, reducing vulnerabilities by 30%
- Enhanced database management and data handling logic to support new service integrations and features, boosting scalability

PROJECTS

AgroSure (Infosys TechForGood - Finalist)

FastAPI, ReactJS, NodeJS, Groq, MongoDB, Gemini, Google Cloud, Firebase, GitHub

Dec. 2024

- An AI-powered web application that caters the Indian farmers
- A crop score is assigned to individual farmers according to their owned land area, location and type of crop.
- The crop score is used to replace their CIBIL score to ease the provision of loans from banks and insurance claims.
- Personalized RAG enabled chatbot - **KisaanSaathi** trained with tailored FAQs
- Custom voice-input enabled IVR system for individuals with low or no accessibility to technology

ShramikSaathi (StatusCode 2.0 - Winner)

ReactJS, Firebase, Llama, Google cloud platform, MongoDB, microservices, GitHub

Jul. 2025

- A web platform for made specifically for the worker-class individuals.
- IVR enabled marketplace for workers to get jobs and users to post jobs as well.
- Community forum, Loan application based on past work history, RAG enabled customized chat agent for specific queries and tailored solutions.

TECHNICAL SKILLS

Languages: Python, C, Java, SQL, JavaScript, HTML, CSS

Frameworks: ReactJS, NodeJS, FastAPI, REST API, TensorFlow, PyTorch, keras, Scikit-learn

Developer Tools: GitHub, Google Cloud Platform, Firebase, HuggingFace, Vercel, Netlify, VS Code, Cloudflare, Spline3D, Framer motion, PyCharm, Android Studio **Libraries:** pandas, NumPy, Matplotlib, OpenCV

CERTIFICATIONS

Advance system security topics - *University of colorado*

Information theory - *University of Hong kong*

Introduction to machine learning - *IIT Madras*

ROLES AND ACHIEVEMENTS

Webmaster - IEEE Computer Society, IEM

Junior research fellow - IEM IEDC, InnovAI